3. PREREQUISITES

Development Environment

- **Node.js (v16+) & npm/yarn:** Required to build and run the Vite + React frontend environment.
- **Python (3.10+):** Required to run the FastAPI backend, along with `pip` for package management.
- **Web Browser:** A modern browser (Chrome, Firefox, Safari) with LocalStorage enabled to support state persistence across sessions.

Cloud Services & API Keys

- **Groq Cloud API Key:** A valid API key is strictly required to facilitate LLaMA multimodal inference and reasoning.
- **Cloud Hosting Accounts (Optional for Deployment):** An account with Vercel (for frontend hosting) and Render/Railway (for backend hosting).

Technologies & Frameworks

- **ReactJS →** https://react.dev
- **FastAPI →** https://fastapi.tiangolo.com
- **OpenCV →** https://opencv.org
- **PyTorch →** https://pytorch.org
- **Meta LLaMA Models →** https://ai.meta.com/llama
- **Python 3.9+ →** https://python.org
- **Visual Studio Code →** https://code.visualstudio.com